



PAYAM POURYOUSEFI

✉ payampouryousefi@gmail.com

🌐 [LinkedIn](#)

🐙 [GitHub](#)

EDUCATION

M.Sc | Computer Science

GPA: 19.22/20 (1st in class) | Nour Azad University

Creating a Benchmark for Evaluating Optimization and Learning Algorithms in Factorio

2021 – 2025

Mazandaran, Iran

Dr. Kourosh Nemati

B.Eng | Electrical Engineering

GPA: 13.29/20 | Shahid Beheshti University

Development of Closed-Form Equations for Design of Metamaterial Unit-Cells

2012 – 2017

Tehran, Iran

Dr. Shokrollah Karimian

Diploma | Mathematics and Physics

Shahed High School

GPA: 19.67/20

2008 – 2012

Chaloos, Iran

RESEARCH INTERESTS

- Machine Learning
- Deep Learning
- Benchmarking
- Robotics
- Reinforcement Learning
- Generative Models
- Bioinformatics
- Genetic Programming
- Evolutionary Computation
- Scientific Computing
- Explainable/Fair AI
- Operational Research

RESEARCH EXPERIENCE

Graduate Researcher | Machine Learning Benchmark Development

In conjunction with the Banzhaf lab - Michigan State University

Supervisors: Dr. Iliya Miralavy - Dr. Kenneth Reid

2022 – 2025

Remote

Electrical Engineering Lab | Microcontroller Programming and Circuit Design

Shahid Beheshti University

Supervisor: Dr. Omid Hashemipour

2014 – 2015

Tehran, Iran

AWARDS AND HONORS

1st in Class

Ranked 1st in M.Sc. Computer Science class (GPA: 19.22/20), Nour Azad University

2025

Rank 59

in national university entrance exam for Master's program in Computer Science

2020

Top 0.2%

in national university entrance exam for Bachelor's program in Mathematics and Science

2012

Mathematics and Olympiad Team Member

Mazandaran and Chaloos teams, including 1st Place Math Team in a National Scientific Competition

2009 – 2011

PUBLICATION

- Payam Pouryousefi, Iliya Miralavy, Kourosh Nemati, Ali Tavassoli, Kenneth N. Reid, and Wolfgang Banzhaf. "Factorio Benchmark: A Machine Learning Toolkit". (In preparation, 2025).
- Milad Mohammadnejad, Saber Rahmatian, Payam Pouryousefi, Mojtaba Rasouli Gharehviran. "A Computational Study of Heat Transfer Enhancement Inside a Channel Using Vortex Generation by Rotating Rotors" (in Farsi), in Proc. International Conference on Modern Approaches to Engineering Sciences (ICMAES01), Tbilisi, Georgia, (2019). Available: <https://en.civildica.com/doc/814793/>
- Salar Radman, Sajjad Movaffaq, Payam Pouryousefi. "The Effect of the Overlap Ratio of the Blades on the Performance of Savonius Wind Turbine, in the Batchtype Models", in Proc. Fourth International Congress of Engineering, Technology and Applied Sciences, Auckland, New Zealand, (2019). Available: <https://isnac.ir/XBFC-DEZDZ>

PROFESSIONAL EXPERIENCE

Play Tester	2022 – 2023
MCDM Productions - Evaluated game mechanics and provided feedback on balance and user experience	
System Designer	2022
Space Sucks - Designed core systems, collaborated on game mechanics, and cross-system integration	
Junior Programmer	2020 – 2021
Ariady - Implemented UI migration, designed business logic, & contributed to technical business planning	

PROJECTS COMPLETED

Mathematical Modeling of Agent Population Dynamics	2023
Simulated differential equation models to analyze periodic agent population dynamics	
PI Controller Parameter Tuning	2023
Identified PI controller parameters using optimization-based system identification	
Factorio Benchmark (FaBe)	2022 – current
Developed a ML benchmark in Factorio video game; evaluated using GA, PSO, and GP algorithms	
Derivative-Free Optimization	2022
Implemented and compared Gravitational Search and Genetic Algorithms for optimization problems	
Multi-dimensional K-means Clustering	2021
Implemented a clustering algorithm for high-dimensional data analysis	
MIPS 5-Stage Pipeline in VHDL	2016
Designed and implemented a pipelined MIPS processor architecture	

TEACHING EXPERIENCE

Advisor on Completion of Bachelor Final Year Project on Load Balancers in Factorio	2024 – 2025
Coburg University	Online
Programming with Python	2022 – 2024
Private tutor	Tehran, Iran
High-school Mathematics and Science	2018 – 2021
Private tutor	Chaloos, Iran
English	2017 – 2018
Pardis English institute	Chaloos, Iran

SKILLS

TECHNICAL SKILLS

Programming: Python, Kotlin, C++

Machine Learning and Visualization Libraries: DEAP, Numpy, SciPy, Matplotlib, Pandas

Development Tools: PyCharm, Visual Studio Code, Sublime Text, Anaconda (Spyder), IntelliJ IDEA

Version Control: Git, GitLab

Familiar with: Docker, LaTeX, Lua, C, MATLAB, Miro, Sphinx

Language: English: C1 (IELTS: 8)

SOFT SKILLS

- | | | |
|-------------------------------|------------------------------|--------------------------|
| • Critical Thinking | • Creativity | • Adaptability |
| • International Communication | • Analytical Problem Solving | • Collaborative Research |
| • Empowering Leadership | • Cultural Competence | • Emotional Intelligence |